Online Appendix 3. Additional Results

This appendix is associated with Witteman and colleagues, 2021, Clarifying Values: An Updated and Expanded Systematic Review and Meta-Analysis.

Additional Meta-analytic Results

Figure S1. Risk of Values-Incongruent Decisions by Tradeoffs

This analysis compares the risk of values-incongruent decisions for values clarification methods that explicitly require the user to engage with tradeoffs in some way versus those that do not require such engagement.

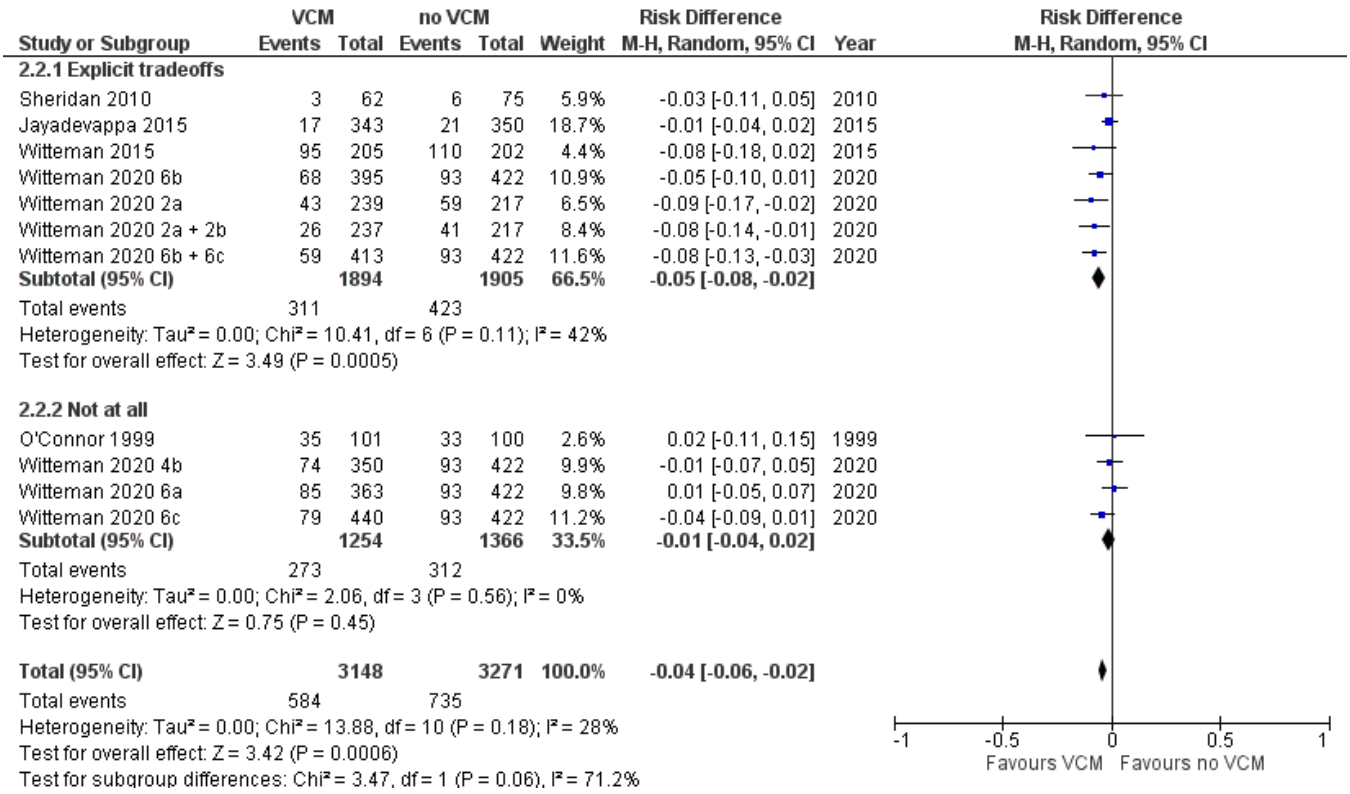
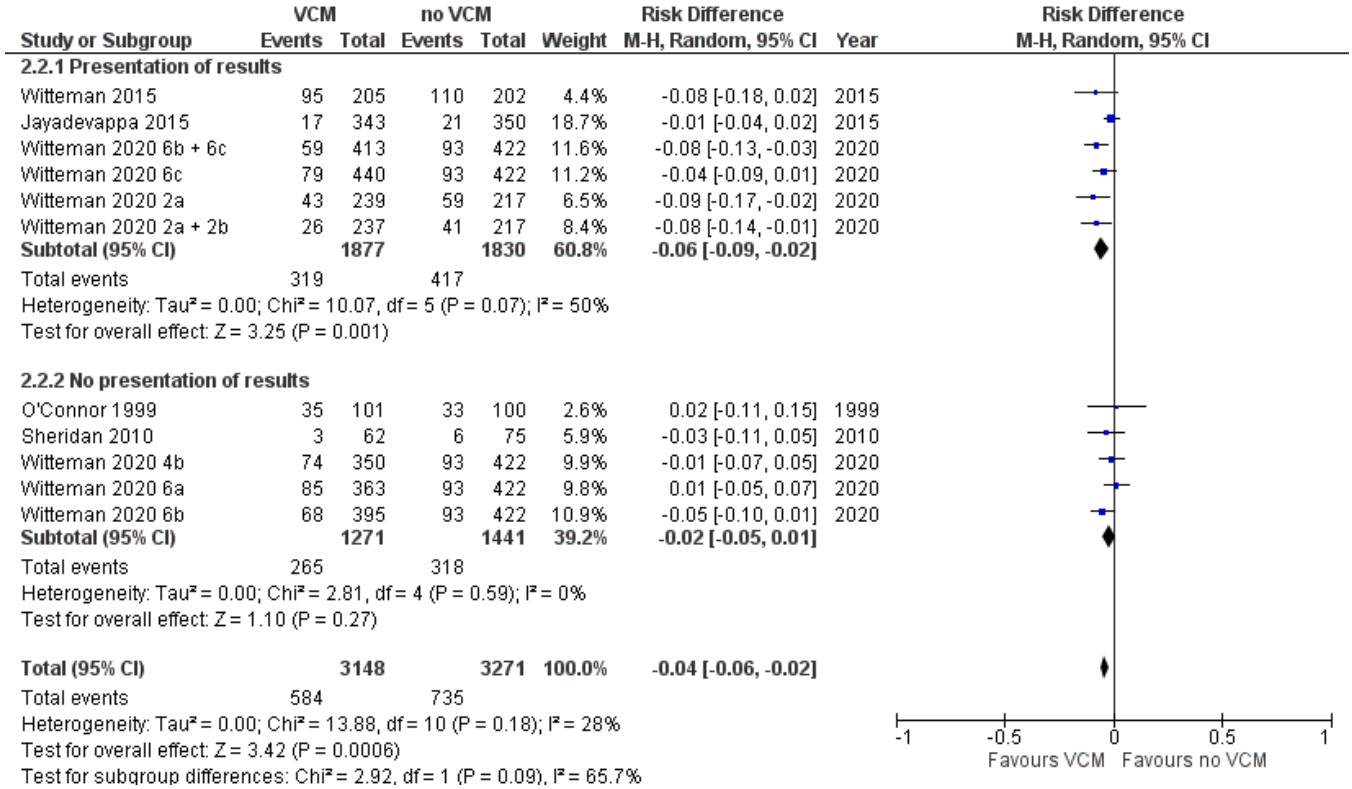


Figure S2. Risk of Values-Incongruent Decisions by Implications/Presentation of Results

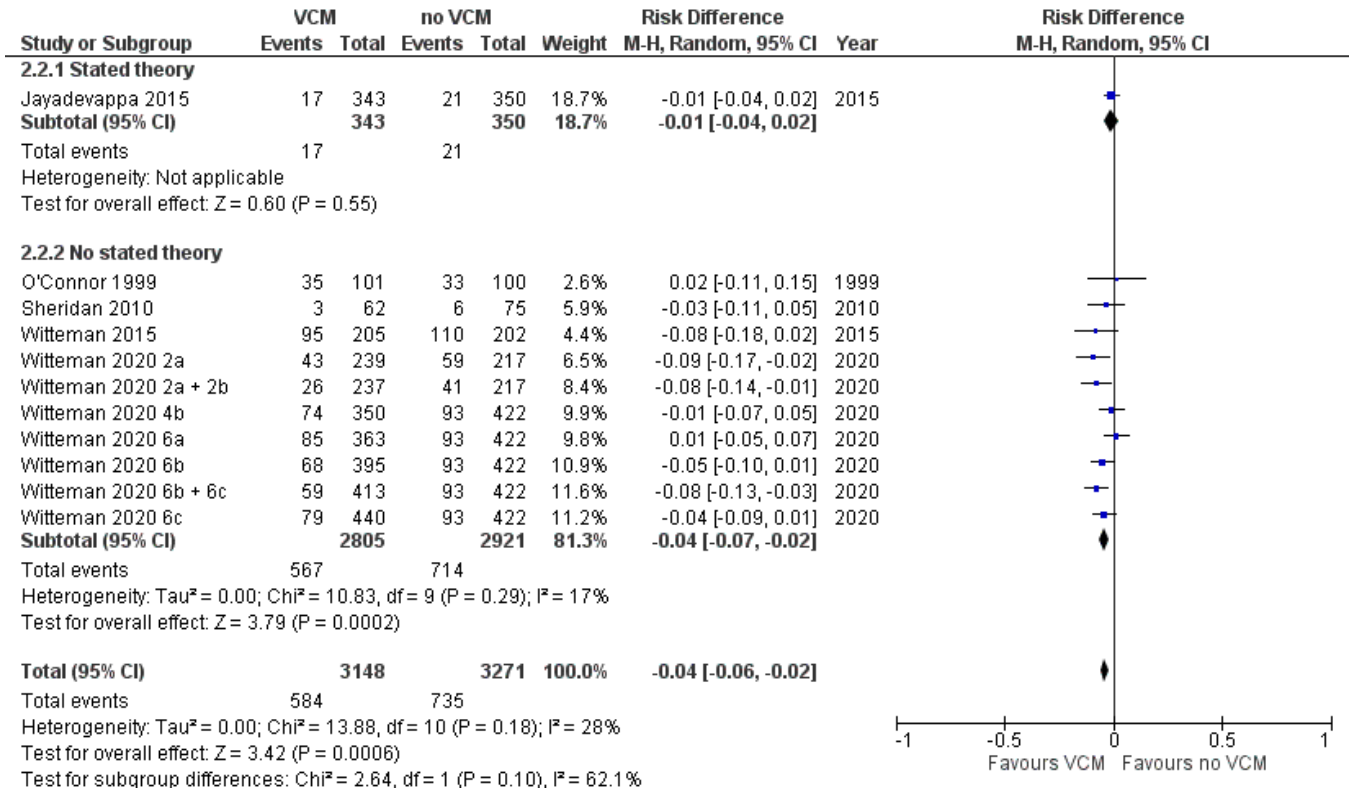
This analysis compares the risk of values-incongruent decisions for values clarification methods that explicitly show the user with the implications of what they value (i.e., present the user with the 'results' of their values clarification process) versus those that do not show the user any such thing.



Study or Subgroup	VCM Events	VCM Total	no VCM Events	no VCM Total	Weight	Risk Difference M-H, Random, 95% CI	Year
2.2.1 Presentation of results							
Witteman 2015	95	205	110	202	4.4%	-0.08 [-0.18, 0.02]	2015
Jayadevappa 2015	17	343	21	350	18.7%	-0.01 [-0.04, 0.02]	2015
Witteman 2020 6b + 6c	59	413	93	422	11.6%	-0.08 [-0.13, -0.03]	2020
Witteman 2020 6c	79	440	93	422	11.2%	-0.04 [-0.09, 0.01]	2020
Witteman 2020 2a	43	239	59	217	6.5%	-0.09 [-0.17, -0.02]	2020
Witteman 2020 2a + 2b	26	237	41	217	8.4%	-0.08 [-0.14, -0.01]	2020
Subtotal (95% CI)		**1877**		**1830**	**60.8%**	**-0.06 [-0.09, -0.02]**	
Total events	319		417				
Heterogeneity: Tau² = 0.00; Chi² = 10.07, df = 5 (P = 0.07); I² = 50%							
Test for overall effect: Z = 3.25 (P = 0.001)							
2.2.2 No presentation of results							
O'Connor 1999	35	101	33	100	2.6%	0.02 [-0.11, 0.15]	1999
Sheridan 2010	3	62	6	75	5.9%	-0.03 [-0.11, 0.05]	2010
Witteman 2020 4b	74	350	93	422	9.9%	-0.01 [-0.07, 0.05]	2020
Witteman 2020 6a	85	363	93	422	9.8%	0.01 [-0.05, 0.07]	2020
Witteman 2020 6b	68	395	93	422	10.9%	-0.05 [-0.10, 0.01]	2020
Subtotal (95% CI)		**1271**		**1441**	**39.2%**	**-0.02 [-0.05, 0.01]**	
Total events	265		318				
Heterogeneity: Tau² = 0.00; Chi² = 2.81, df = 4 (P = 0.59); I² = 0%							
Test for overall effect: Z = 1.10 (P = 0.27)							
Total (95% CI)		**3148**		**3271**	**100.0%**	**-0.04 [-0.06, -0.02]**	
Total events	584		735				
Heterogeneity: Tau² = 0.00; Chi² = 13.88, df = 10 (P = 0.18); I² = 28%							
Test for overall effect: Z = 3.42 (P = 0.0006)							
Test for subgroup differences: Chi² = 2.92, df = 1 (P = 0.09), I² = 65.7%							

Figure S3. Risk of Values-Incongruent Decisions by Stated Use of a Theory/Framework

This analysis compares the risk of values-incongruent decisions for values clarification methods that are underpinned by a formal theory or conceptual framework versus those that are not.



Study or Subgroup	VCM Events	VCM Total	no VCM Events	no VCM Total	Weight	Risk Difference M-H, Random, 95% CI	Year
2.2.1 Stated theory							
Jayadevappa 2015	17	343	21	350	18.7%	-0.01 [-0.04, 0.02]	2015
Subtotal (95% CI)		**343**		**350**	**18.7%**	**-0.01 [-0.04, 0.02]**	
Total events	17		21				
Heterogeneity: Not applicable							
Test for overall effect: Z = 0.60 (P = 0.55)							
2.2.2 No stated theory							
O'Connor 1999	35	101	33	100	2.6%	0.02 [-0.11, 0.15]	1999
Sheridan 2010	3	62	6	75	5.9%	-0.03 [-0.11, 0.05]	2010
Witteman 2015	95	205	110	202	4.4%	-0.08 [-0.18, 0.02]	2015
Witteman 2020 2a	43	239	59	217	6.5%	-0.09 [-0.17, -0.02]	2020
Witteman 2020 2a + 2b	26	237	41	217	8.4%	-0.08 [-0.14, -0.01]	2020
Witteman 2020 4b	74	350	93	422	9.9%	-0.01 [-0.07, 0.05]	2020
Witteman 2020 6a	85	363	93	422	9.8%	0.01 [-0.05, 0.07]	2020
Witteman 2020 6b	68	395	93	422	10.9%	-0.05 [-0.10, 0.01]	2020
Witteman 2020 6b + 6c	59	413	93	422	11.6%	-0.08 [-0.13, -0.03]	2020
Witteman 2020 6c	79	440	93	422	11.2%	-0.04 [-0.09, 0.01]	2020
Subtotal (95% CI)		**2805**		**2921**	**81.3%**	**-0.04 [-0.07, -0.02]**	
Total events	567		714				
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 10.83$, df = 9 (P = 0.29); $I^2 = 17\%$							
Test for overall effect: Z = 3.79 (P = 0.0002)							
Total (95% CI)		**3148**		**3271**	**100.0%**	**-0.04 [-0.06, -0.02]**	
Total events	584		735				
Heterogeneity: $Tau^2 = 0.00$; $Chi^2 = 13.88$, df = 10 (P = 0.18); $I^2 = 28\%$							
Test for overall effect: Z = 3.42 (P = 0.0006)							
Test for subgroup differences: $Chi^2 = 2.64$, df = 1 (P = 0.10), $I^2 = 62.1\%$							

Figure S4. Risk of Values-Incongruent Decisions by Real/Hypothetical Decision

This analysis compares the risk of values-incongruent decisions for values clarification methods tested in the context of real decisions versus those tested in the context of hypothetical decisions.

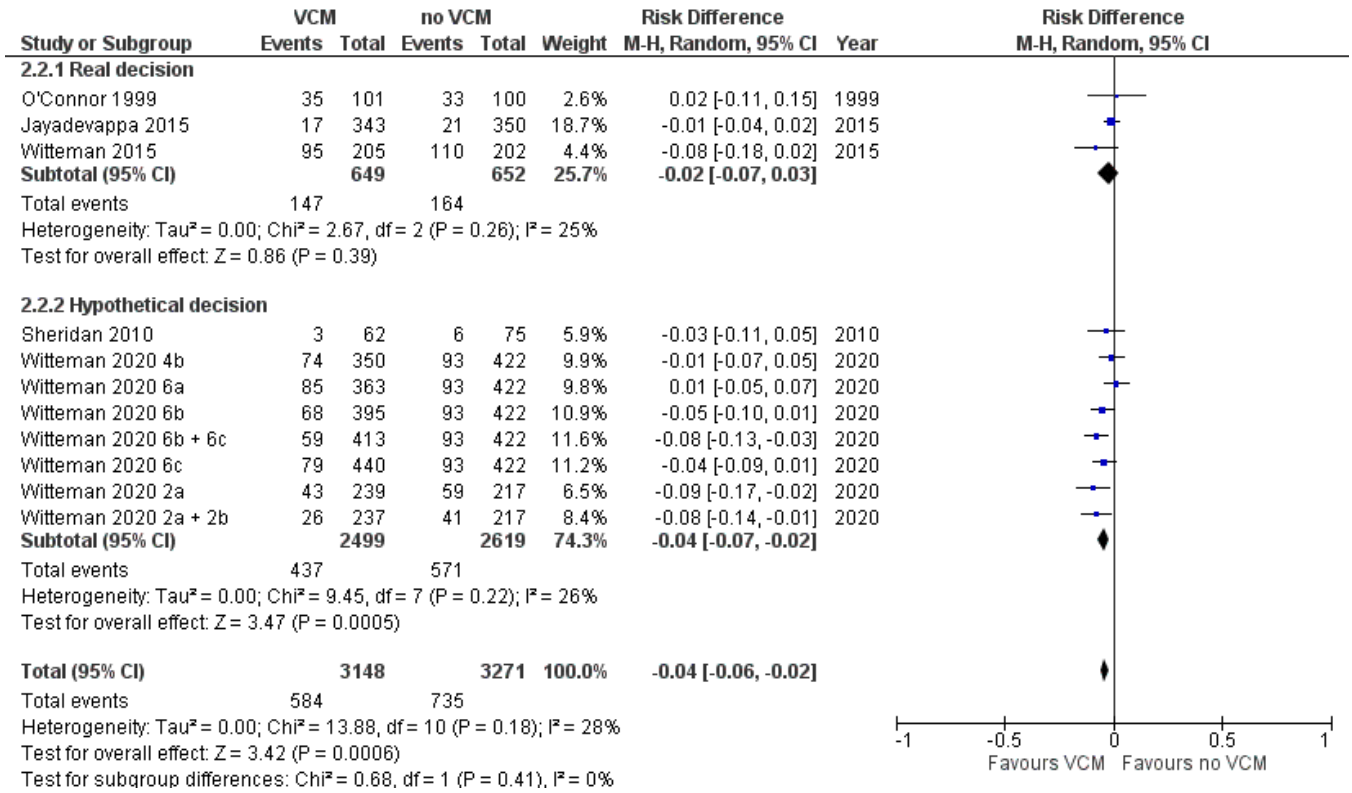


Figure S5. Decisional Conflict by Tradeoffs

This analysis compares decisional conflict for values clarification methods that explicitly require the user to engage with tradeoffs in some way versus those that do not require such engagement.

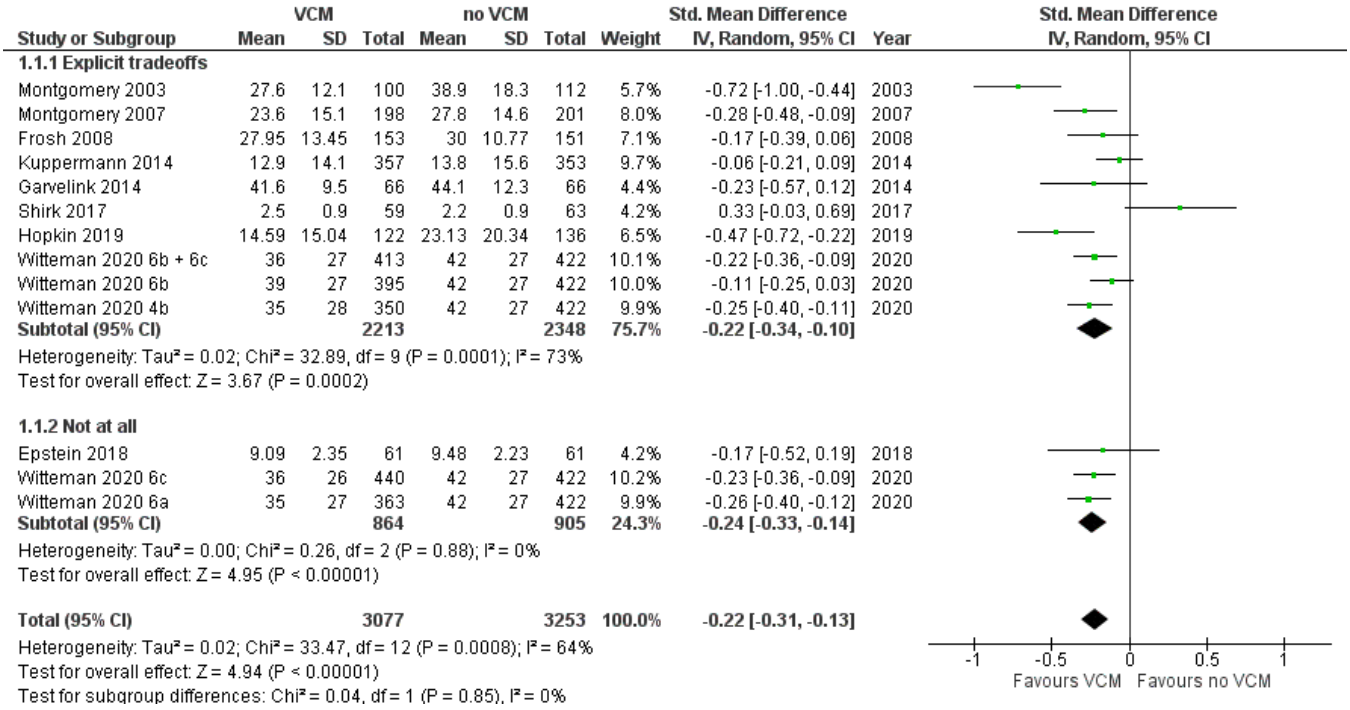


Figure S6. Decisional Conflict by Implications/Presentation of Results

This analysis compares decisional conflict for values clarification methods that explicitly show the user with the implications of what they value (i.e., present the user with the 'results' of their values clarification process) versus those that do not show the user any such thing.

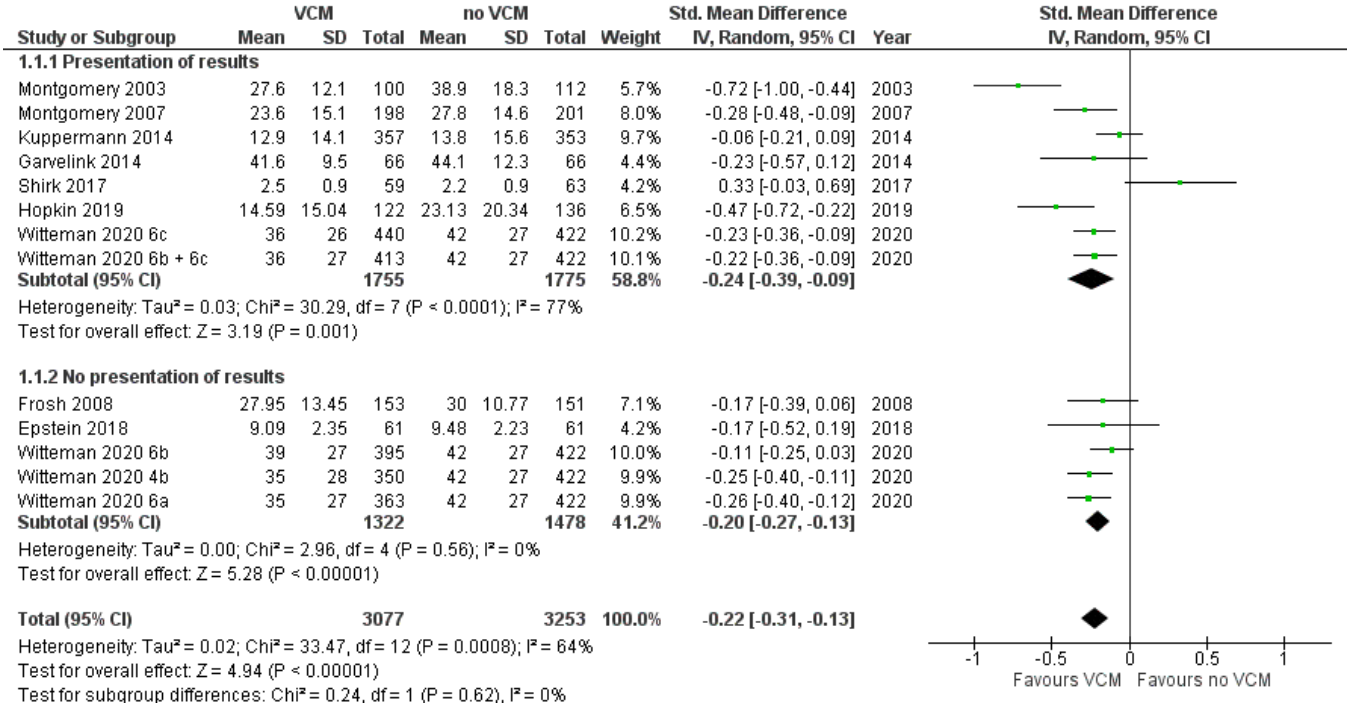


Figure S7. Decisional Conflict by Stated Use of a Theory/Framework

This analysis compares decisional conflict for values clarification methods that are underpinned by a formal theory or conceptual framework versus those that are not.

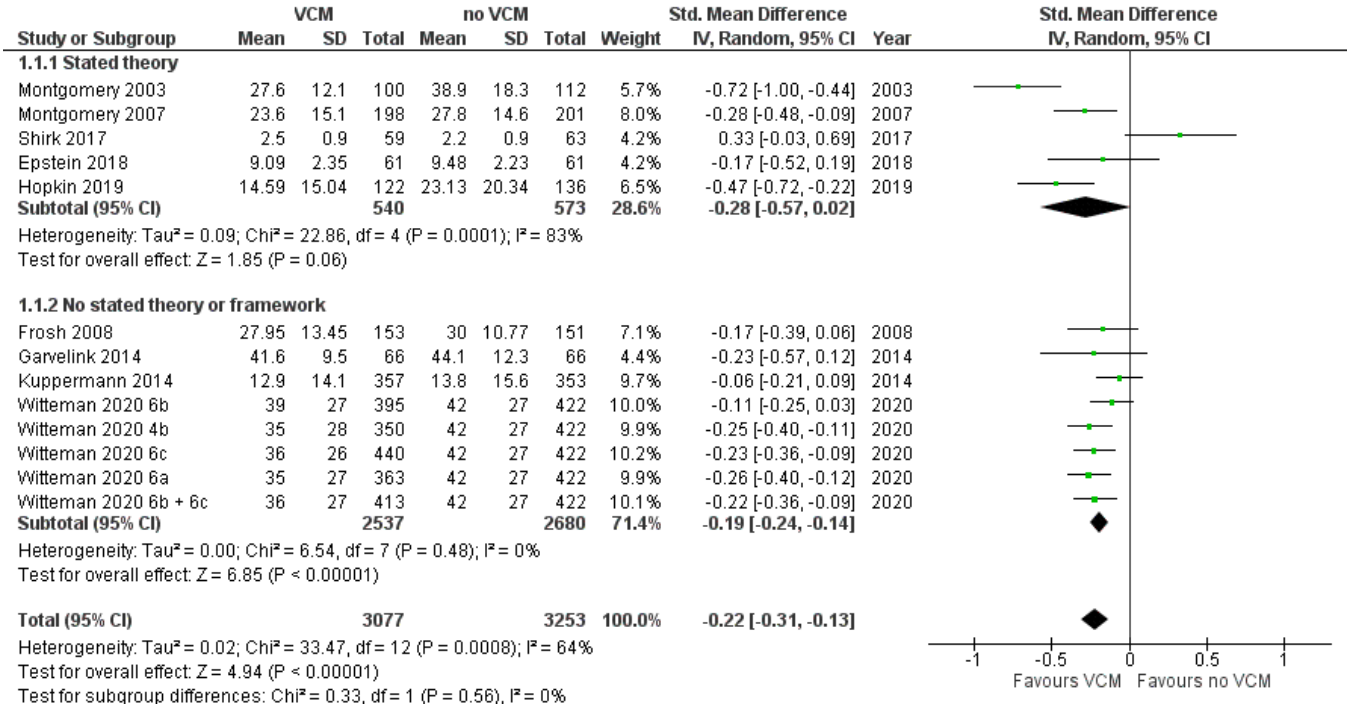


Figure S8. Decisional Conflict by Real/Hypothetical

This analysis compares decisional conflict for values clarification methods tested in the context of real decisions versus those tested in the context of hypothetical decisions.

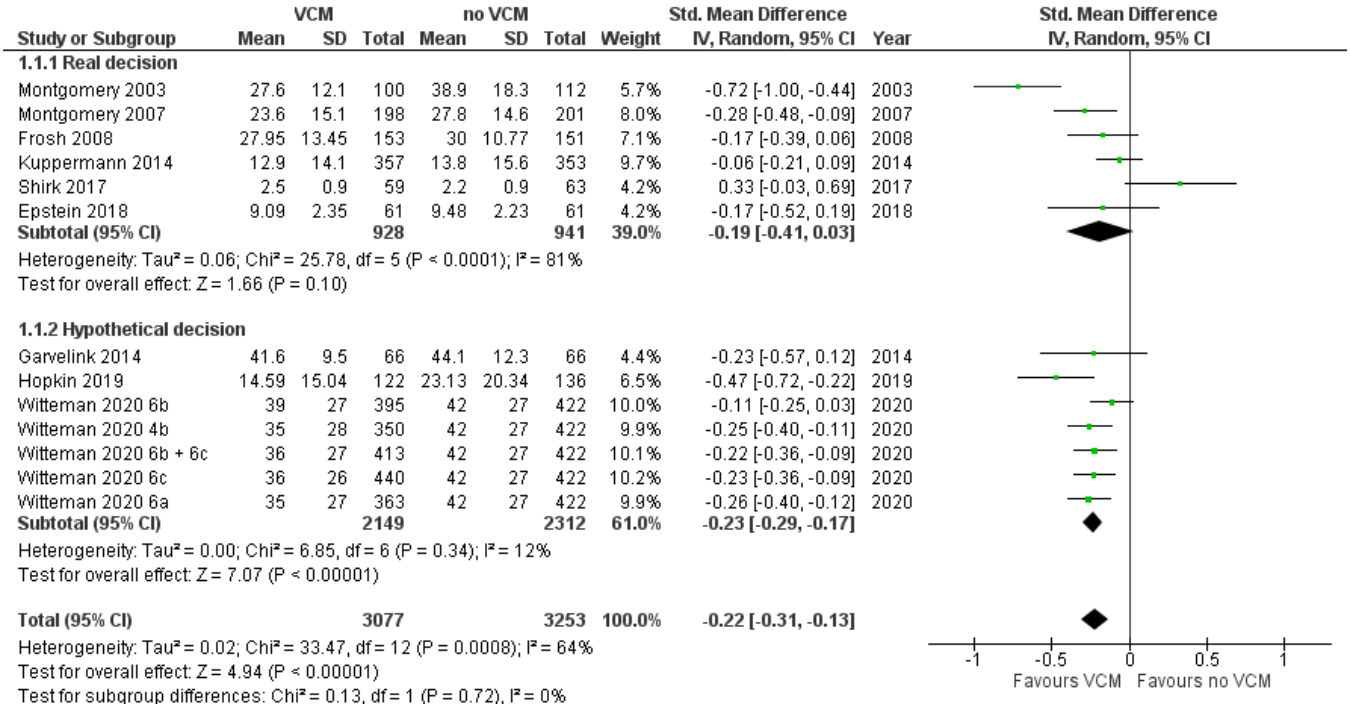
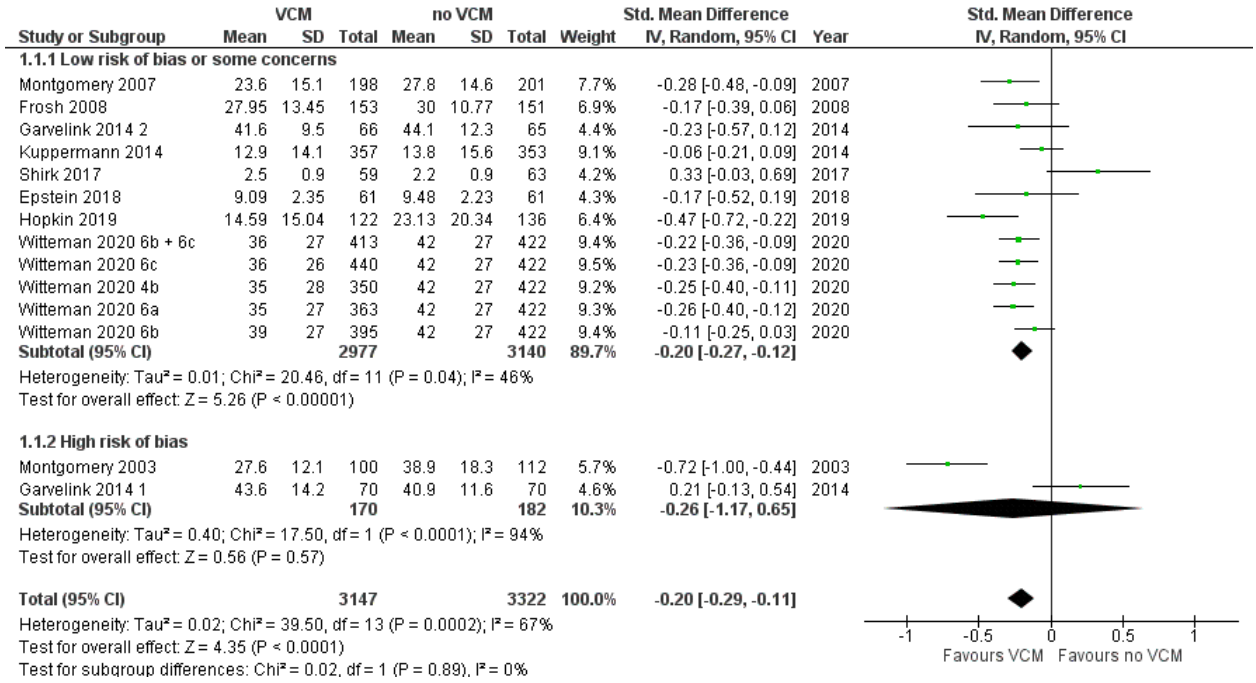


Figure S9. Decisional Conflict by Risk of Bias

This analysis compares decisional conflict for values clarification methods tested in the context of low risk of bias or some concerns versus those tested in the context of high risk of bias. Note that high risk of bias may be necessary in some cases due to the nature of values clarification methods. For example, it may be difficult to conceal the purpose of an intervention from the person using it.



Head-to-Head Evaluations of Values Clarification Methods

Feldman-Stewart et al. (2006) found no difference across all three groups (information only; values clarification method without a summary bar, i.e. rating scales; values clarification with a summary bar, i.e. multicriteria decision analysis) in terms of the attributes participants identified as important to their decisions nor in how difficult it was to make the decision. When trial participants were unblinded at the end of the study and shown all three options, all of them ranked the bars with the summary option (multicriteria decision analysis) as the most helpful.

Pignone et al. (2012) found that a discrete choice experiment produced somewhat different patterns of attribute importance compared to ranking and rating. Agreement between the most important attribute derived from the values clarification method and the most important attribute as reported by participants in the questionnaire was slightly higher in the ranking and rating arm than the discrete choice experiment arm. The authors found no difference between study arms in terms of values clarity, intent to be screened and unlabelled screening test preference.

Pignone et al. (2013) found that different values clarification methods produced differences in attribute importance and screening test preference. Participants who received the rating and ranking test were more likely to report the chance of dying from prostate cancer as the most important attribute compared to the balance sheet and discrete choice experiment groups. Those who received the balance sheet were more likely to prefer the unlabelled PSA-like test option compared to the two other groups. Participants who received the discrete choice experiment were somewhat less likely to select reduction of mortality as the most important attribute, and were least likely to select the PSA-like option on the unlabelled preference question. There was no difference across groups in intent to be screened (labelled PSA test option) nor on values clarity.

Brenner et al. (2014) found that different values clarification methods produced different results in terms of individuals' most important screening test attributes. Specifically, respondents who received the rating and ranking exercise, compared to a discrete choice experiment or a balance sheet (i.e., implicit values clarification method), were the most likely to choose risk reduction as the most important attribute. They found no differences in terms of test preferences, values clarity, nor intention to be screened.

Witteman et al. (2020) found that overall, methods using mathematical models (e.g., decision analysis, allocation of points) were more promising than other methods (e.g., pros and cons, rating scales) for encouraging values-congruent decisions. All methods encouraged lower decisional conflict when this was assessed.

Risk of Bias

Article	Random Sequence Generation (Selection Bias)	Allocation Concealment (Selection Bias)	Blinding of Participants and Personnel (Performance Bias)	Blinding of Outcome Assessment (Detection Bias)	Incomplete Outcome Data (Attrition Data)	Selective Reporting (Reporting Bias)	Other Bias
Abhyankar 2010	low	unclear	high	unclear	low	low	low
Au 2012	low	low	unclear	unclear	unclear	unclear	unclear
Bekker 2004	low	low	unclear	unclear	unclear	low	low
Brenner 2014	low	low	low	unclear	unclear	low	low
Clancy 1988	low	unclear	unclear	low	unclear	unclear	high
deAchaval 2012	low	low	low	low	low	low	low
Epstein 2018	low	unclear	unclear	high	unclear	low	unclear
Feldman-Stewart 2006	unclear	unclear	unclear	unclear	unclear	unclear	low
Feldman-Stewart 2012	low	low	unclear	low	low	unclear	low
Fraenken 2007	low	low	high	unclear	low	low	low
Frosch 2008	low	low	unclear	low	low	unclear	low
Garvelink 2014 (1)	low	low	low	low	unclear	low	high
Garvelink 2014 (2)	low	low	low	low	low	low	unclear
Hess 2015	low	low	unclear	unclear	unclear	low	unclear
Hopkin 2019	low	low	low	low	unclear	low	unclear
Hutyra 2019	low	unclear	low	low	low	low	unclear
Jayadevappa 2019	low	low	unclear	unclear	low	unclear	unclear
Kennedy 2002	low	low	high	unclear	low	unclear	unclear
Kuppermann 2014	low	low	low	low	low	low	unclear
Lehrman 1997	unclear	unclear	high	unclear	low	low	low
Matheis-Kraft 1997	unclear	unclear	unclear	unclear	low	unclear	unclear
Montgomery 2007	low	low	low	low	low	low	unclear
Montgomery 2003	low	low	high	low	low	unclear	low

Myers 2003	unclear	unclear	unclear	unclear	unclear	low	unclear
Myers 2005	unclear	unclear	unclear	unclear	low	low	unclear
O'Connor 1999	low	low	unclear	low	low	unclear	low
Paquin 2018	low	low	unclear	unclear	low	low	unclear
Peinado 2020	unclear	low	unclear	unclear	unclear	low	low
Pignone 2012	unclear	unclear	low	unclear	unclear	low	low
Pignone 2013	low	low	low	unclear	unclear	low	low
Sheridan 2010	low	low	low	low	low	low	low
Shirk 2017	unclear	unclear	unclear	unclear	low	low	unclear
Witteman 2015	low	low	unclear	unclear	unclear	low	unclear
Witteman 2020	unclear	low	unclear	unclear	low	low	low

Sensitivity Analyses of I^2 Estimates

Because Witteman 2020 included two comparisons from one study population (2a; 2a + 2b) and five comparisons from another study population (4b; 6a; 6b; 6c; 6b + 6c), for each outcome, we conducted 10 additional meta-analyses (2 x 5 possible combinations) using a random effects model and 1 additional meta-analysis using a fixed effect model on all comparisons contributed by Witteman 2020. This allowed us to assess I^2 sensitivity to the inclusion of multiple comparisons drawing from the same study populations. Overall, we found that the I^2 estimates were robust to the inclusion of multiple comparisons.

Figure S10. Risk of Values-Incongruent Decisions with Witteman 2020 2a and 4b

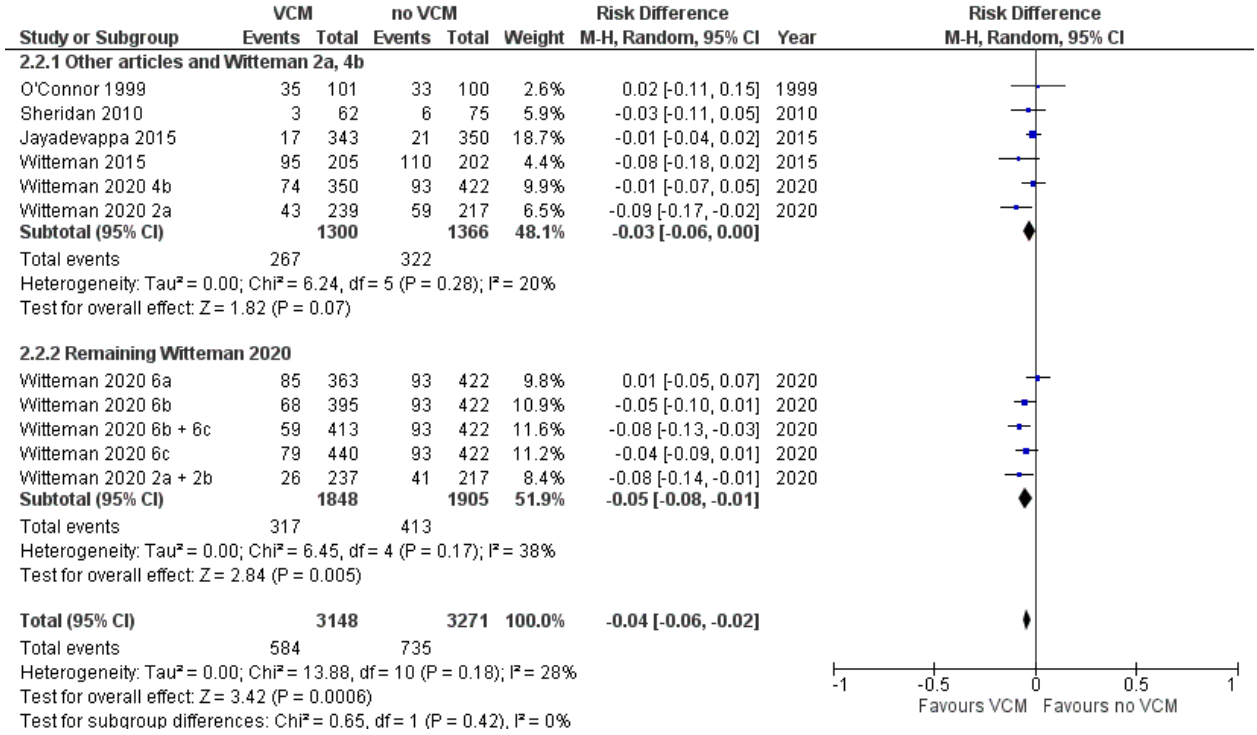


Figure S11. Risk of Values-Incongruent Decisions with Witteman 2020 2a and 6a

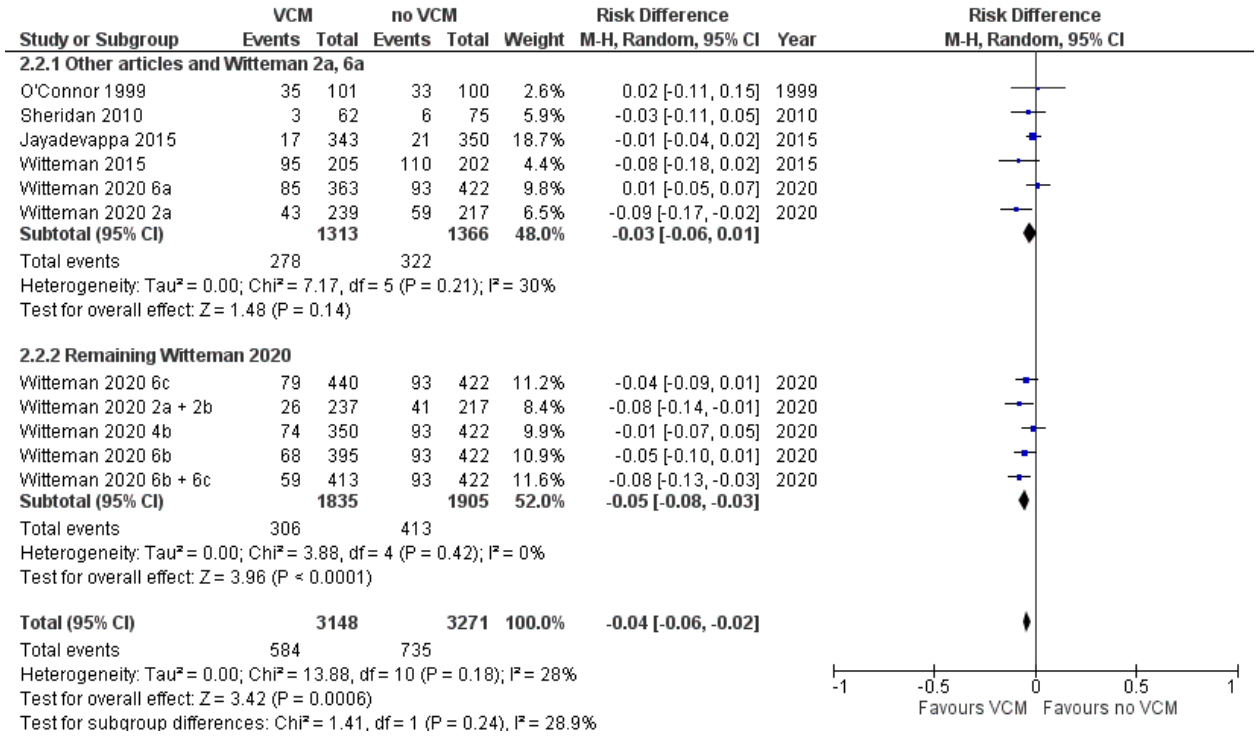


Figure S12. Risk of Values-Incongruent Decisions with Witteman 2020 2a and 6b

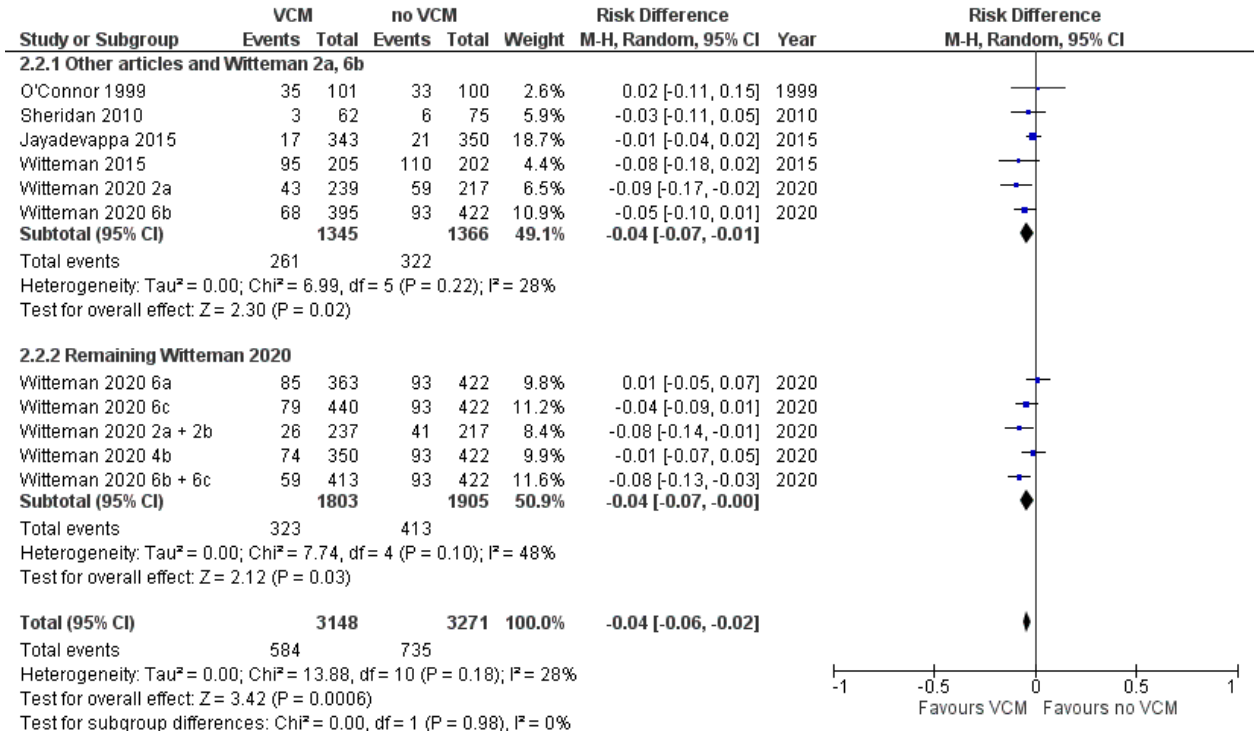


Figure S13. Risk of Values-Incongruent Decisions with Witteman 2020 2a and 6c

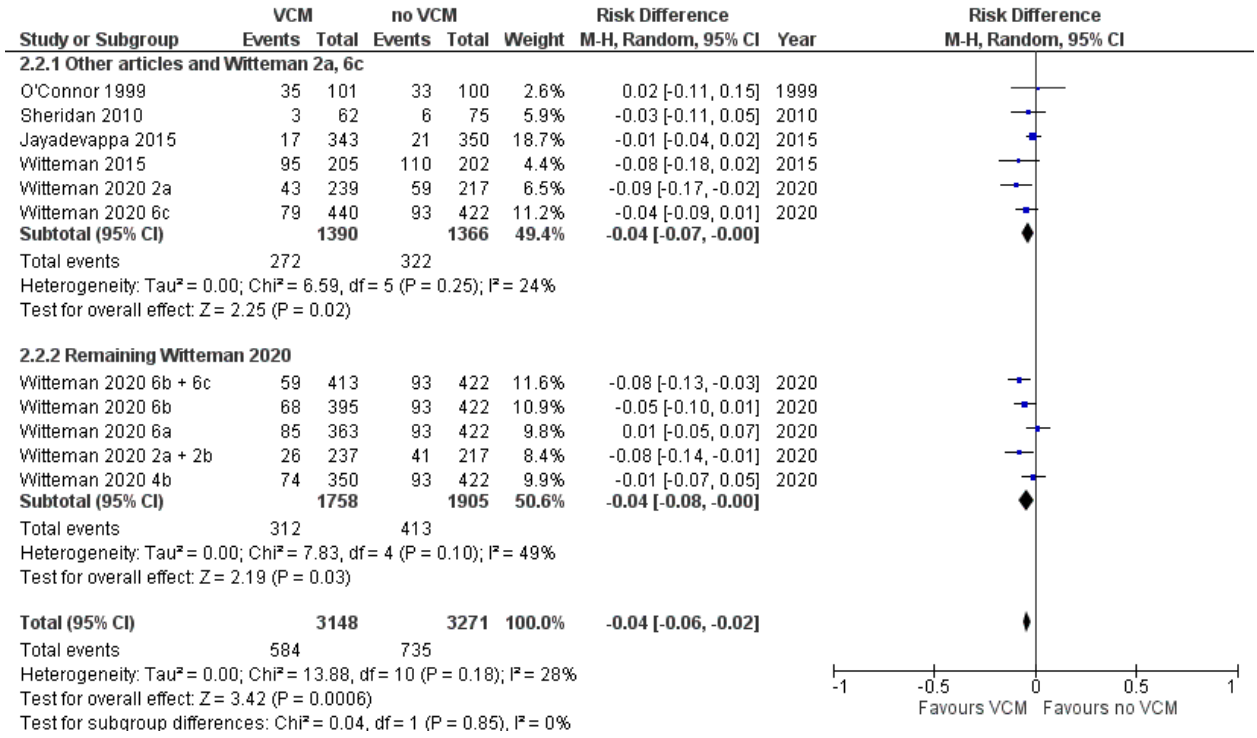


Figure S14. Risk of Values-Incongruent Decisions with Witteman 2020 2a and 6b + 6c

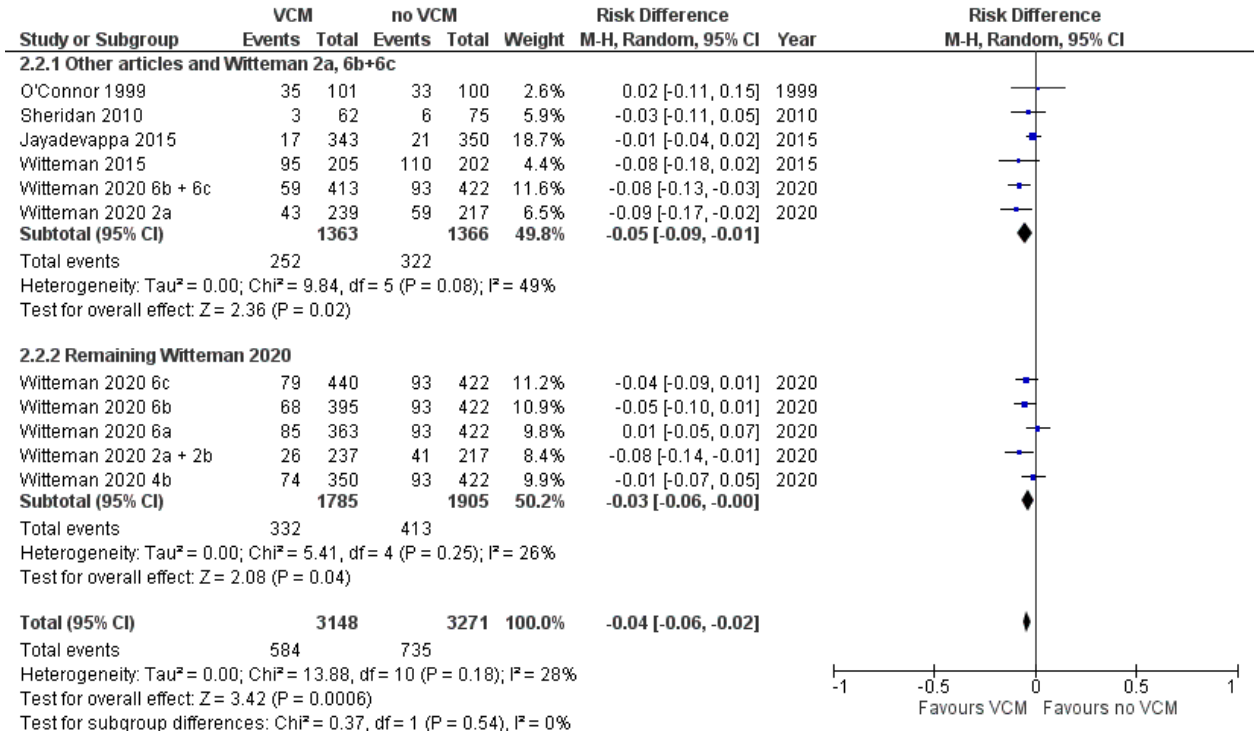


Figure S15. Risk of Values-Incongruent Decisions with Witteman 2020 2a + 2b and 4b

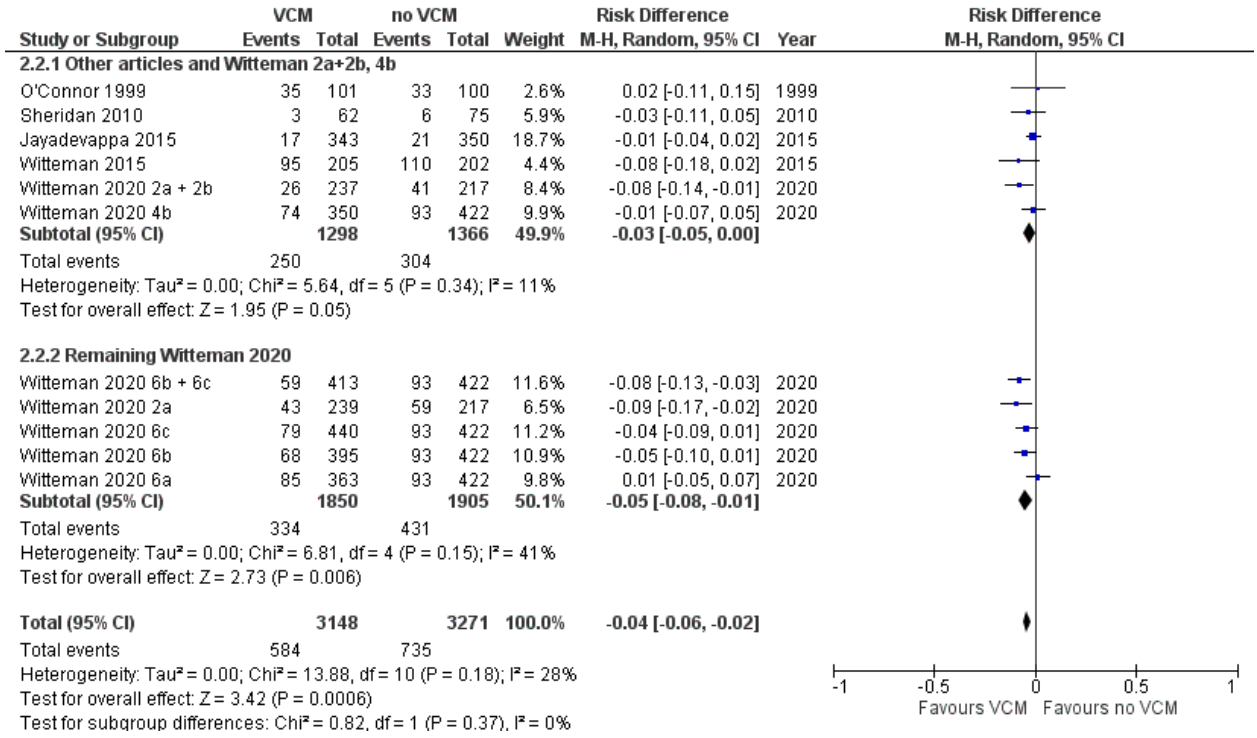


Figure S16. Risk of Values-Incongruent Decisions with Witteman 2020 2a + 2b and 6a

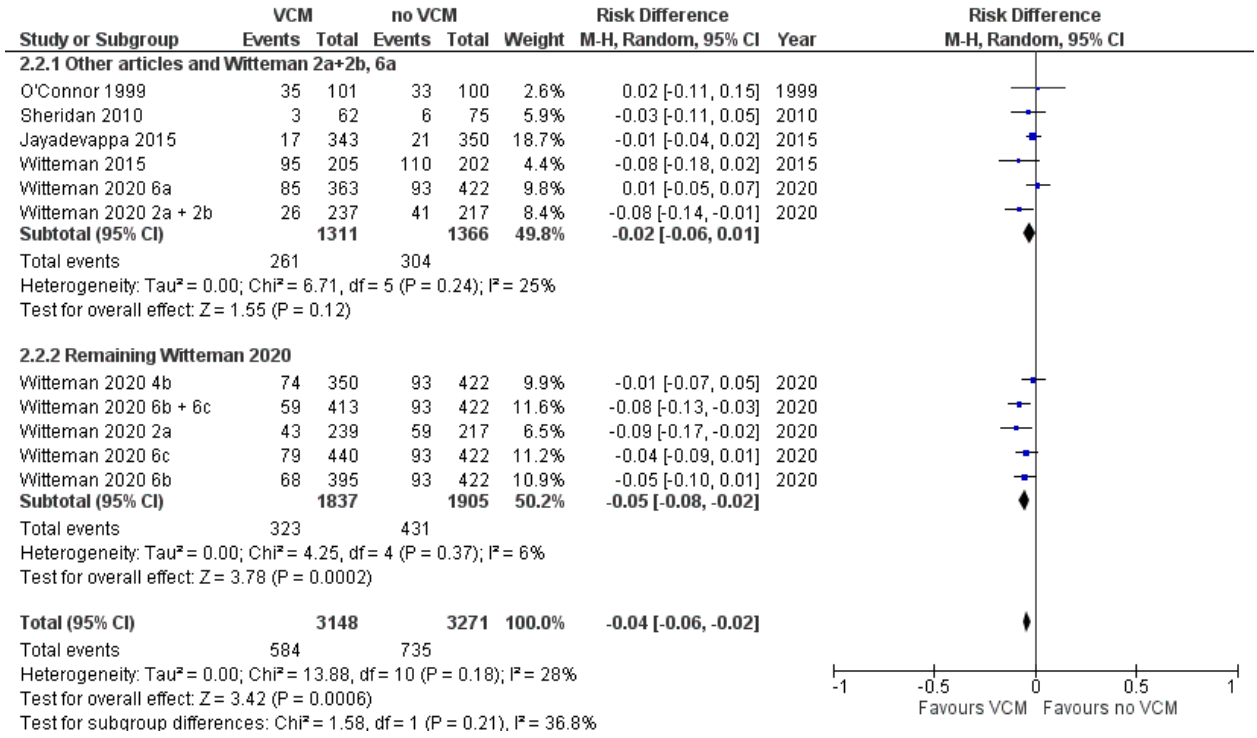


Figure S17. Risk of Values-Incongruent Decisions with Witteman 2020 2a + 2b and 6b

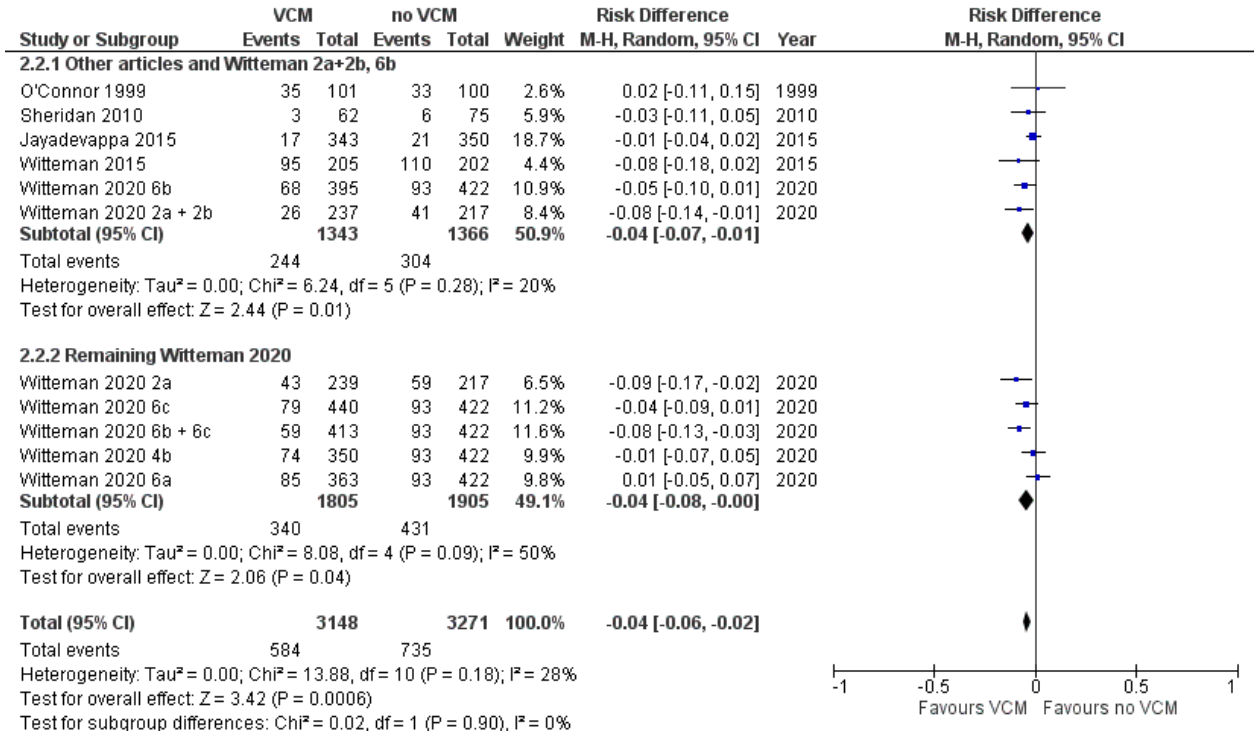


Figure S18. Risk of Values-Incongruent Decisions with Witteman 2020 2a + 2b and 6c

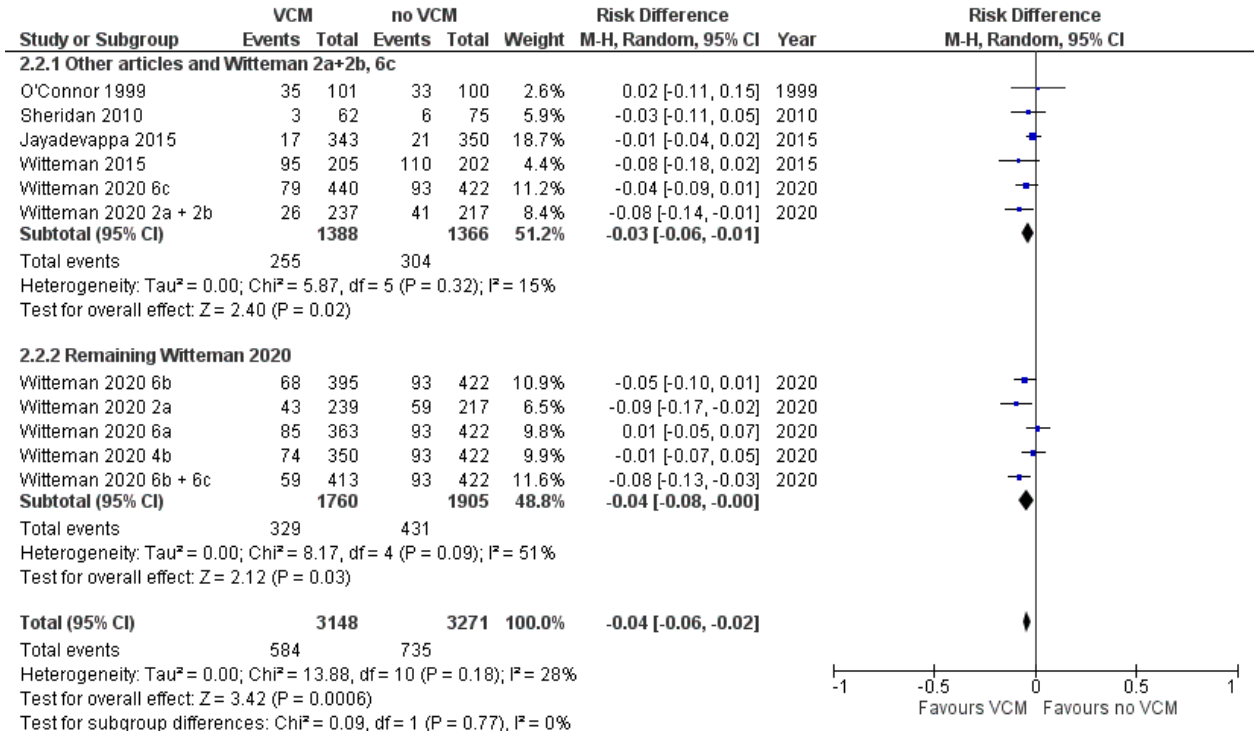


Figure S19. Risk of Values-Incongruent Decisions with Witteman 2020 2a + 2b and 6b + 6c

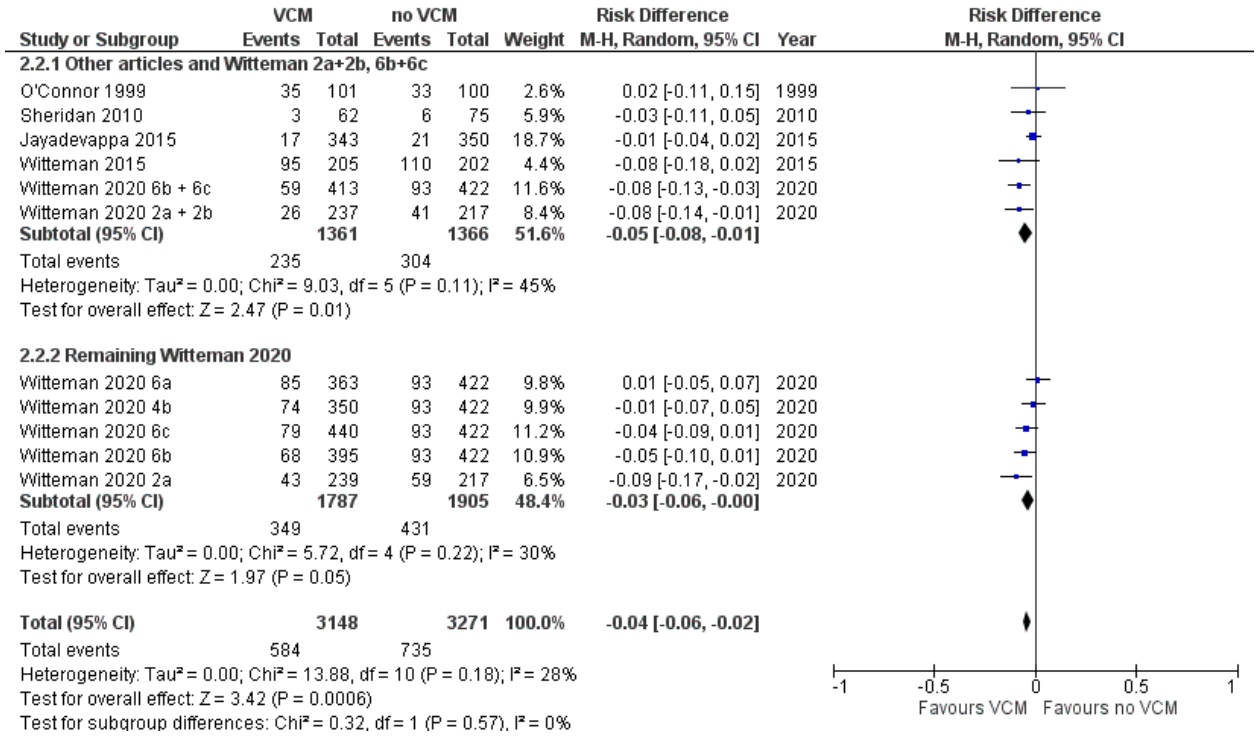


Figure S20. Risk of Values-Incongruent Decisions for Witteman 2020 Studies (Fixed Effects Model)

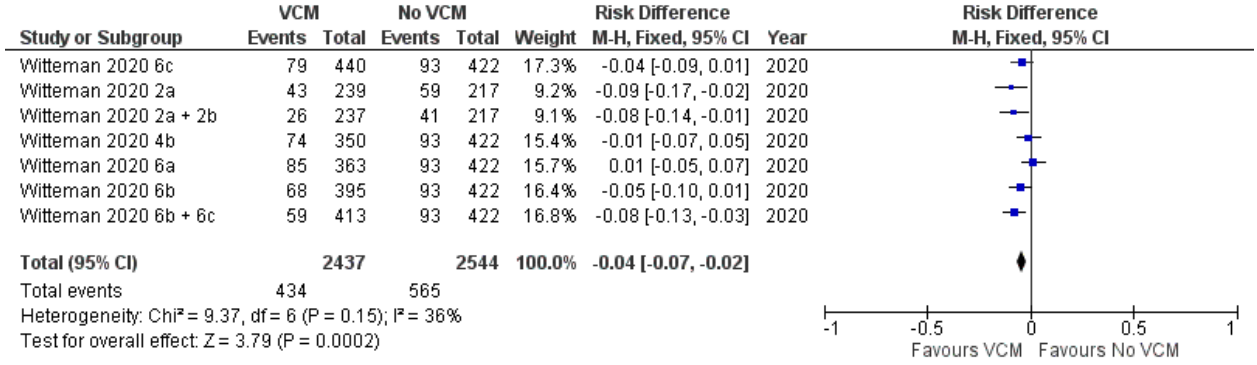


Figure S21. Decisional Conflict with Witteman 2020 4b

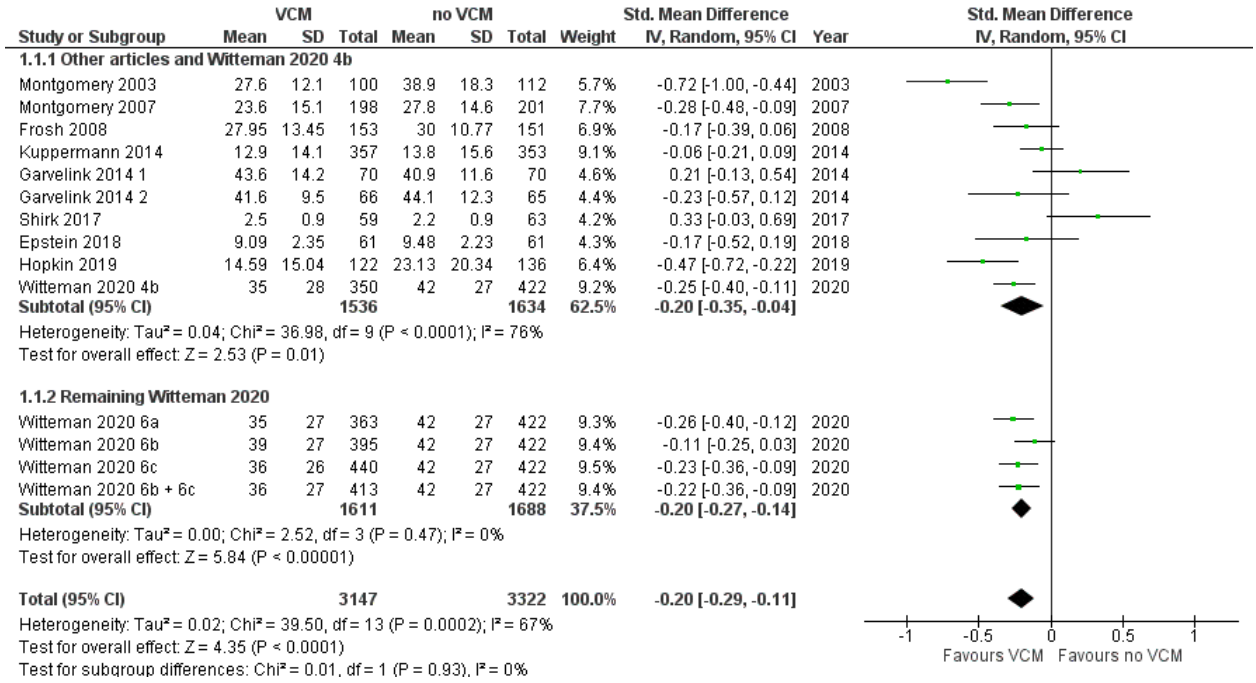


Figure S22. Decisional Conflict with Witteman 2020 6a

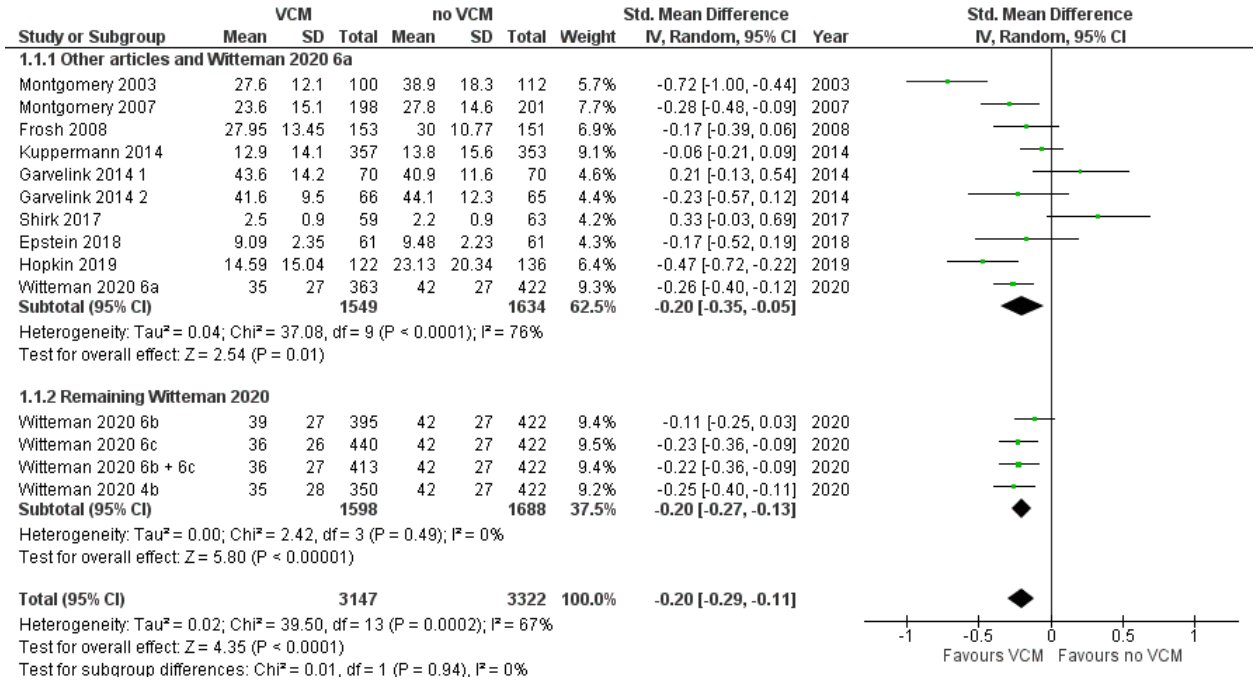


Figure S23. Decisional Conflict with Witteman 2020 6b

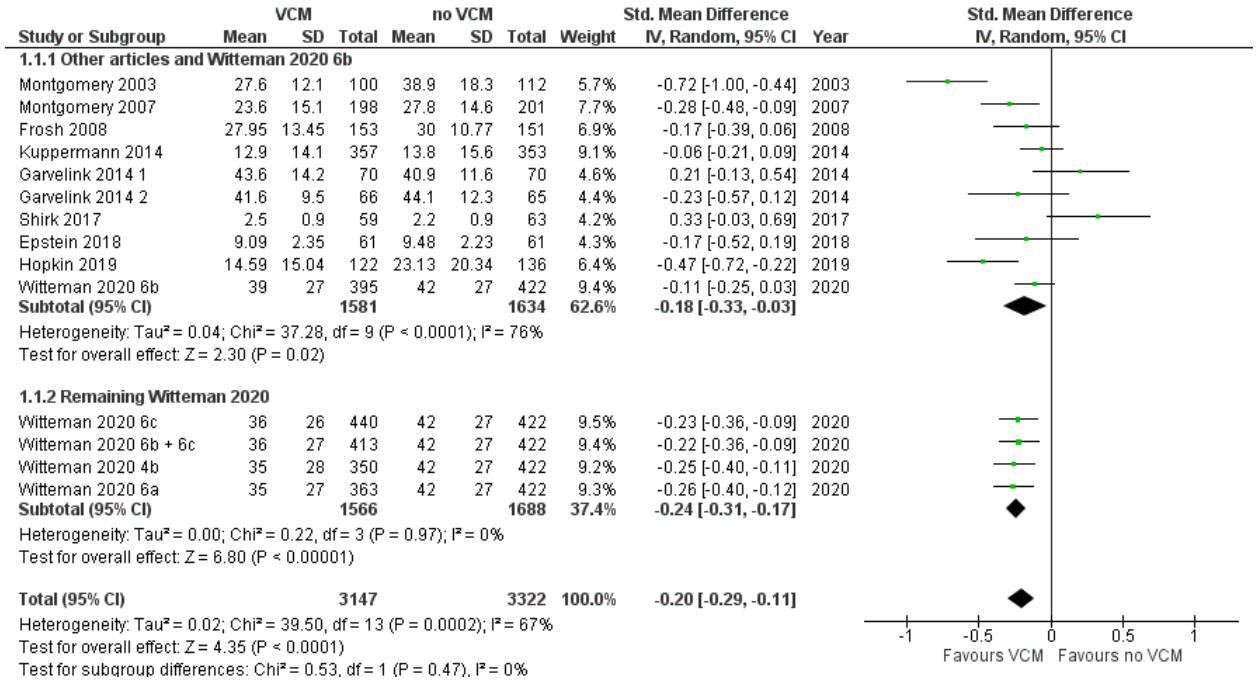


Figure S24. Decisional Conflict with Witteman 2020 6c

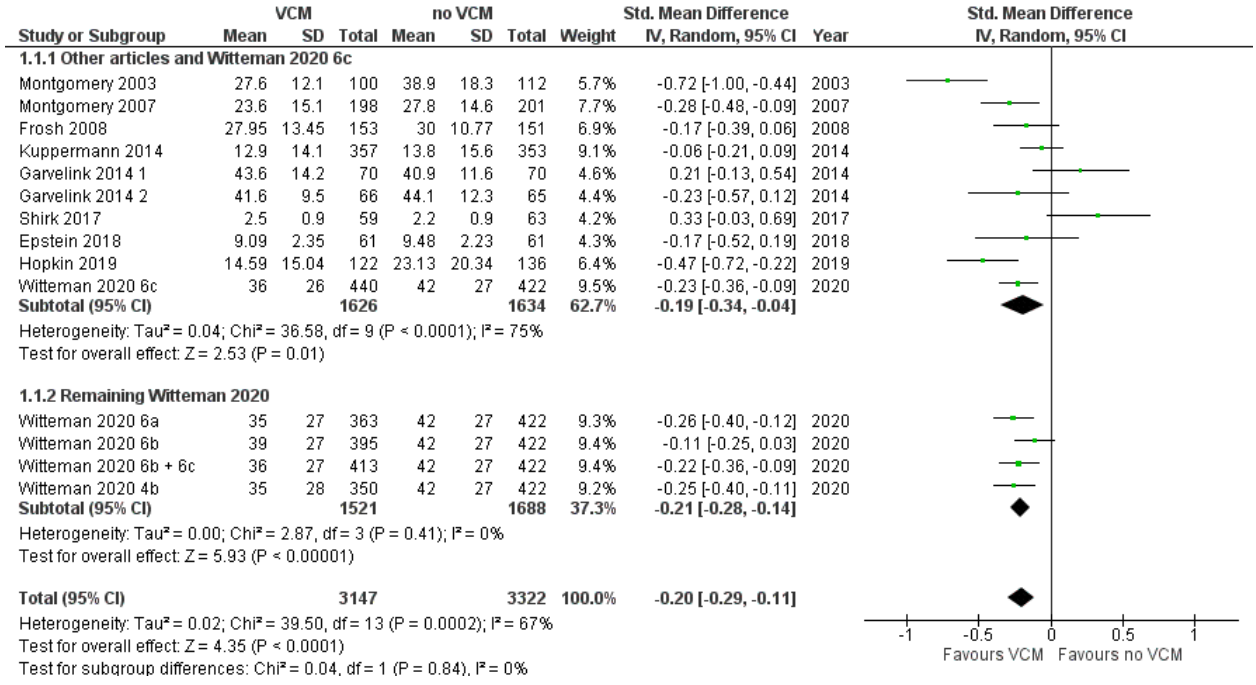


Figure S25. Decisional Conflict with Witteman 2020 6b + 6c

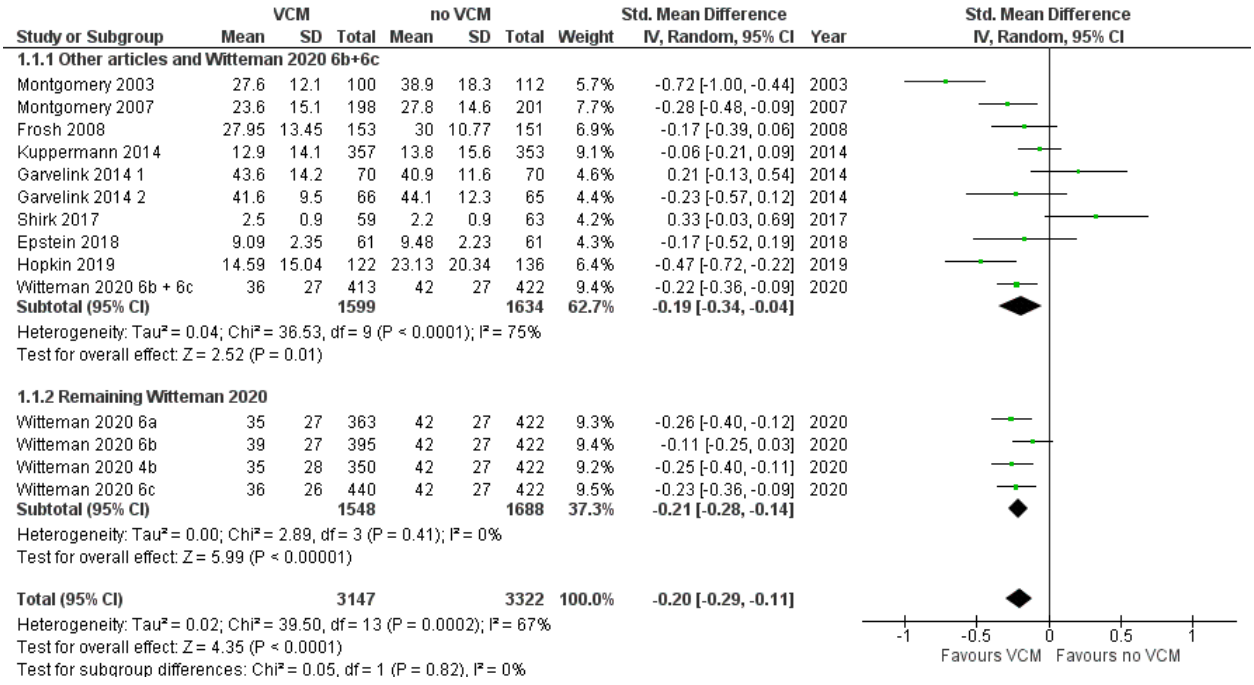
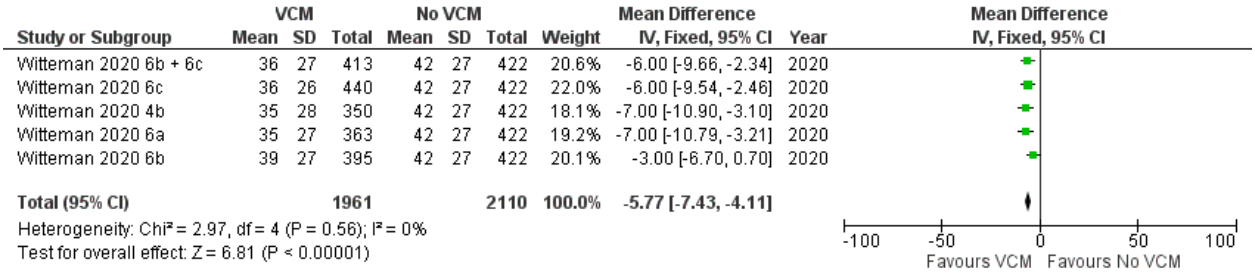



Figure S26. Decisional Conflict for Witteman 2020 Studies (Fixed Effects Model)